CO2 - ASSESSMENT TOOL 3

Course name : Data warehousing and Data mining

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① Define data mining and list any two basic mining tasks. Examplain the steps involved in KDD process and justify the role of data preprossing in it.

Answer:

* Data mining: The process of extracting useful patterns, knowledge, and relationship from large datasets.

* Two basic data mining tasks:

a. classification

b. clustering

* Steps in KDD (Knowledge Discovery in Database):

1. Data cleaning
2. Data Integration
3. Data selection
4. Data Transformation
5. Data mining
6. Pattern evaluation
7. knowledge presentation

* Role of Data preprocessing: It improves data quality by removing errors, handling missing values, integrating multiple sources, and transforming data into a suitable format for mining.

② What is data cleaning? mention two common data quality issues. Explain methods to handle missing values and noisy data ~~preprocessing~~ with suitable examples.

Answer:

Data cleaning is the process of detecting and correcting inaccurate, incomplete, inconsistent, or duplicate data

* Two common data quality issues:

* missing values
* Noisy data.

* Handling missing values:

* Ignore the record
* fill with mean, median, or mode
* predict values using machine learning.

* Handling Nosiy Data:

Binning

Regression

clustering.

Example: Age values: 20, 21, ?, 23 → missing value can be replaced with 21.3 (mean).

③ Define data integration and list two challenges associated with it. example how redundancy and inconsistency are handled during data integration.

Answer:

Data Integration combains data from multiple sources into a single unified dataset.

* challenges:
  1. Data redundancy
  2. Data inconsistency

* Handling Redundancy:
  * Remove duplicate records
  * Use correlation analysis.

* Handling inconsistency:
  * standardize formats.
  * Resolve naming conflicts.
  * Apply data validation rules.

④ what is covariance analysis? list two covariance formals

Answer: Covariance analysis measures how two variables change together.

* Two covariance formula:

population covariance:

$$Cov(x,y) = \frac{\Sigma (x_i - \mu x)(y_i - \mu y)}{N}$$

sample covariance:

$$Cov(x,y) = \frac{\Sigma (x_i - x)(y_i - y)}{n-1}$$

5) Suppose two stocks A and B have the following values in one week : (2,5), (3,8), (5,10), (4,11), (6,14). Question : If the stocks are affected by the same industry trends, will their prices rise or fall together and independent ?

Data : A = {2, 3, 5, 4, 6} B = {5, 8, 10, 11, 14}

Mean (A) = 4

Mean (B) = 9.6

Convariance = 4.5

Answer : since covariance is positive, the prices of stocks A and B rise together (positive relationship). They are not independent.

(6) ~~Example~~ Explain the architure of a typical data mining system with a neat diagram. Discuss the role of each component and analyze how they interact to support knowledge discovery

Data sources.

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Database / Data warehouse server.

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Data cleaning & Integration

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Data mining Engine

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pattern Evaluation

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Knowledge Base

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User Interface

Components :

- Data source : store raw data.
- Database / Data warehouse server : Retrieves data.
- Data mining Engine : finds patterns.
- pattern Evaluation : selects useful patterns.
- knowledge Base : stores domain knowledge
- User Interface : Displays results.

(7) find covariance for following data set $X = \{4, 5, 6, 7, 9\}$,
$Y = \{8, 3, 7, 5, 6\}$.

A) Given : $X = \{4, 5, 6, 7, 9\}$

$Y = \{8, 3, 7, 5, 6\}$

mean $(X) = 6.2$

mean $(Y) = 5.8$

sample covariance :

$$\text{Cov} = \frac{\Sigma (X - \bar{X})(Y - \bar{Y})}{4} = \frac{-2.8}{4} = -0.7$$

Covariance $= 0.7$

(8) Describe the major steps in data pre-processing. Given a dataset with missing values, outliers, and inconsistent formats, propose a systematic pre-processing pipeline and justify the sequence of steps.

1. Data Cleaning
2. Data Integration
3. Data Transformation
4. Data Reduction
5. Data Discretization.

Pipeline :

1. Handle missing values.
2. Remove outliers.
3. Standardize data formats
4. Normalize data
5. Reduce dimensions if needed.
6. Discretize continuous values.

Reason: This order improves data quality before mining and produces more accurate results.

9) A hospital is analyzing patient age data to identify broad helth trends. Because the row data contains minor variations and noise, the analyst needs to perform local smoothing using equal-frequency (equi-depth) binning. sample Data set (Ages):

13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70

Data: 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 22, 25, 25, 25, 30, 33, 33, 35, 35, 35, 36, 40, 45, 46, 52, 70.

Using 3 equal-frequency bins (9 values each):

Bin 1: 13, 15, 16, 16, 19, 20, 20, 21, 22

Bin 2: 22, 25, 25, 25, 25, 30, 33, 33, 35

Bin 3: 35, 35, 35, 36, 40, 45, 46, 52, 70

Answer: Equal-frequency binning smooths local noise while keeping approximately the same number values in each bin.

(10) Explain data discretization and its importance and its dat minins. Compare equal-width and equal-frequency binning with suitable example, and evaluate their impact on data analysis

* Data Discretization

Definition: Data discretization converts continous numerical values into intervals (bins).

* Importance:

- Reduce data complexity.
- improves mining efficiency
- makes patterns easier to understand

equal-width Binning	equal-frequency Binning
* Equal interval size	* Equal number of records
* simple to implement	* Better for uneven data
* may produce uneven bins	* Balances data across bins.

Example : Data : 10, 20, 30, 40, 50, 60

* Equal-width : (10-26), (27-43), (44-60)
* Equal-frequency : (10, 20), (30, 40), (50, 60).

Impact :

→ Equal-width is suitable for uniformly distributed data

→ Equal-frequency performs better on skewed datasets.